

local alternative for public cloud storage, which mitigates the cost and risks associated with public cloud storage. This network attached and cloud storage platform is quite suitable for small to medium businesses as well as home users who do not have much IT experience but need to scale up to terabytes of data storage. If users are more interested in Linux and Btrfs, it is a great alternative to FreeNAS. This cloud storage platform can be managed even within a LAN or over the Web using a very simple and intuitive user interface. And with the inclusion of add-ons (named 'Rockons'), you can extend the feature set to include different new applications, servers and services.

**Kinetic Open Storage:** Backed by different companies like Seagate, EMC, Toshiba, Cisco, Red Hat, NetApp, Dell, etc, Kinetic is a Linux Foundation project which is dedicated to establishing standards for new kinds of object storage architecture. It is designed especially to meet the need for scale-out storage used for unstructured data. Kinetic is basically a way for storage applications to communicate directly with storage devices over the Ethernet. Most of the storage use cases targeted by Kinetic consist of unstructured data like Hadoop, NoSQL and other distributed file systems, as well as object stores in the cloud such as Amazon S3, Basho's Riak and OpenStack Swift.

**Storj DriveShare and MetaDisk:** Storj is a new type of cloud storage which is built on peer-to-peer and blockchain technology. It offers decentralised and end-to-end encrypted cloud storage. The DriveShare application allows users to rent out all their unused hard drive space so that it can be used by the service. The MetaDisk Web application present in it allows users to save all their files to the service securely. The core protocol helps in peer-to-peer negotiation and verification of the storage contracts. Providers of the storage are usually referred to as 'farmers' and those using the storage are called 'renters'. Renters can periodically audit in order to check if the farmers are still keeping their files secure and safe. Conversely, farmers can also decide to stop storing any specific file if its owners do not pay and audit their services on time. Different files are cut up into smaller pieces called 'shards' and then are

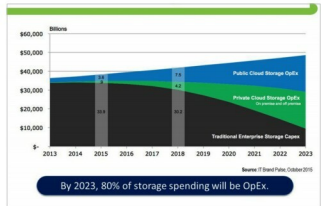


Figure 4: Ten year Data centre revenue forecast (Image source: [googleimages.com](http://googleimages.com))

stored three times redundantly, by default. The network can automatically determine a new farmer and can also move data if copies become unavailable. The system puts different measures in place to prevent renters and farmers from cheating on each other—for instance, by manipulating the auditing process. Storj offers several advantages over many traditional cloud based storage solutions. As data present here is encrypted and cut into shards at the source, there is almost no chance for any unauthorised third parties to access the data. And because data storage is distributed, the availability and download speed increases. 

## References

- [1] <http://www.wikipedia.org/>
- [2] <http://novipro.com/>
- [3] <https://thesanguly.com/>
- [4] <https://www.techrepublic.com>

## By: Vivek Ratan

The author has completed his B.Tech in electronics and instrumentation engineering. He is currently working as an automation test engineer at Infosys, Pune and as a freelance educator at LearnerKul, Pune. He can be reached at [ratavivek14@gmail.com](mailto:ratavivek14@gmail.com).

THE COMPLETE MAGAZINE ON OPEN SOURCE

**OpenSource**  
ForYou

The latest from the Open Source world is here.

**OpenSourceForU.com**

Join the community at [facebook.com/opensourceforu](https://facebook.com/opensourceforu)

Follow us on Twitter @OpenSourceForU